The answer is 50,005,000, by the way.

And what about the ridiculously large number?

(6,516,516,546,516,546,519,654,872,346,987,287,364,598,712,634,578,926,347,856,243,578 + 1) x 6,516,516,546,516,546,519,654,872,346,987,287,364,598,712,634,578,926,347,856,243,578 / 2

It's too big for a standard calculator on your PC, but using an online large number precision calculator, you can find that the answer is:

21,232,493,950,511,969,000,243,100,012,144,292,075,004,317,854,393,293,183,111,910,806,178,703,480,438,260,397,827,087,431,343,781,778,350,404,940,909,789,272,361,242,831

Even a computer would struggle to do this in the 1+2+3+4+5+6... way, and yet here we were able to do it so easily, all because we broke things down into simple math problems that computers can understand.

If you were to calculate this for, say, the sum of numbers between 4 and 10, this would be $(y^2/2 + y/2) - [(x-1)^2/2 + (x-1)/2]$, where x is the smaller number (4) and y is the larger number (10). This can be further simplified into: $(y^2-x^2+x+y)/2$

... since $y^2 - x^2 = (y+x)*(y-x)$, we can further simplify the equation to:

$(y-x+1)*(y+x)/2$

And now you (and a computer) can find the sum of consecutive numbers between any two values of x and y!

Not every problem is easily translated to math though, but this is no problem for a computer, because you just need to use variables, or placeholders in order to get the machine to understand it. The great thing about computers is that they're dumb, so they don't need to understand things in order to run simulations or process algorithms. Let's say you're building a game, and want to make a randomised face generator. Of course, you would have to deconstruct a face for a computer to understand. The more you deconstruct, the simpler you make it and the more variables you have (desirable in a random generator). In a very simplistic example, let's say you were building a game with cartoon cats (everyone loves cats in videos and games, not so much in real life). The first thing you'd do to generate cartoon cat faces is to decide on how many different shapes of the head you want. Because they're cartoons, you can be as creative as you want, and have, say, square, round, oval-wide and oval-tall as four simple shapes (you could add more), next you tell the computer that every cat has two ears, and you model different types of ears, and also show the computer where to place the ears for each head shape. This is just x-y location data, since this is not 3D. Next, a nose, then the same thing for whiskers, mouth-open and mouth closed features, etc.

Obviously, the more parameters you add the more variables you have and the more things you have to specify for each variable. Next, you give the computer individual images, and when it generates the image it puts it together and you have a randomly generated cat face. Sounds simple, but it's a lot of hard work breaking down a cat's face into elements and then trying to experiment with placement of the components until you get it right. An easy cheat would be to ensure that eyes and ears and mouth can all be placed in the same spot, regardless of head shape, which would make your life much easier. Of course, this is just an example of how you could break up something abstract like cat faces into simpler problems that can be solved. Your own use would require you to do that for a problem you have, and the more creative you are with the abstraction, the better the output, and it's this abstraction that computers find it hard to do. They're getting better though. Head on over to thispersondoesnotexist.com and you will find a computer generated image of a person who, as the name suggests, does not exist. These faces are created by what's called a generative adversarial network (GAN). A GAN is essentially two algorithms competing to try and win a game, and that game is to please you and give you what you ask of it. It's like supercharged tinkering and natural selection all rolled into one. Both algorithms come up with an answer, and either you can choose who did better, or you can even automate that by creating a discriminative network that does that job for you! Of course we're oversimplifying here, a lot, but the work has all been done for you already and open sourced, so your level of interaction need not be any more advanced than we're suggesting... of course if you are a coding whiz, you will obviously do a better job at tweaking to suit your requirements and thus achieve better results.

At thispersondoesnotexist.com, computers are generating faces by studying a vast library of faces, and being very careful to never use all the features of anyone's real face in a single image, and thus recreating a fake person. Some of the faces look strikingly real, while others just feel wrong somehow, but overall it's an impressive job. You can even use the source code to build your own generator. All you really need is a large sample of images and you're good to go.

GO FISH!

This month's cover story is to get you thinking along the lines of a tinkerer about everything, and to get you to learn to achieve better results. No geek is ever happy with anything the way it is, so we're sure there are a million things you'd like to tweak. In the articles that follow you will find less abstract and more practical ways to tweak your browser, learn how to code, tweak Android to suit your needs, and even use voice assistants to achieve tasks. All of those will pale in comparison to the complexity of your life however, and it would take us a thousand articles to hack all aspects of one person's life using the tricks we've mentioned. Multiply that by a million readers, and we'd have to write a billion articles just to cover the lives of our readers alone!

As the old saying goes, give a man a fish and you feed him for a day, teach a man to fish and feed him for a lifetime...

Make sure to write in and tell us how you have used tinkering and computational thinking in your life, and we'd love to share it with other readers, so that they get more ideas. And if you have created (or do create) any interesting generators with GAN, tell us and we'd love to feature you in future articles.